

SMART MOSQUITO CONTROL SYSTEM

Project ID: 25-26J-369

Project Proposal Report

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**SMART MOSQUITO CONTROL SYSTEM –
MOBILE/WEB HEALTH DATA INTEGRATION
PLATFORM**

25 – 26J – 369

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date

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Date

Abstract

Increasing prevalence of mosquito-borne disease in Sri Lanka demands a transition towards proactive instead of reactive public health intervention. This project phase reflects the acute demand for accurate, location-specific forecasting in a vision to innovate and implement a robust Machine Learning (ML) prediction model. The aim is to utilize historical mosquito density datasets obtained from the Gampaha MOH area to identify complex breeding patterns and correlations with environmental variables. Data are processed using Python-based data science methodologies to clean, normalize, and structure the raw government records for algorithmic analysis. Historical trends will be modeled using predictive algorithms (such as Regression or Time-Series forecasting), converting static past records into dynamic future density projections. Validation metrics will be generated to benchmark the model's accuracy against actual occurrences, providing actionable intelligence for health authorities to facilitate targeted resource allocation. Implementation of this study will bring a reality of a new, predictive analytics capability at the heart of the general goal of reducing vector-borne transmission of disease in the Gampaha district.

Table of Contents

Contents

Declaration.....	3
Abstract	4
1.Introduction.....	6
2. Background and Literature Survey.....	7
3. Research Gap	7
4. Research Problem	8
5. Objectives	8
5.1 Main Goal:	Error! Bookmark not defined.
5.2 Specific Goals:.....	Error! Bookmark not defined.
6. Methodology	9
6.1 Requirements Analysis ; Technology Selection.....	Error! Bookmark not defined.
6.2 Prototyping & Unit Testing	Error! Bookmark not defined.
6.3 System Integration & Testing	Error! Bookmark not defined.
6.4 Field Testing & Rollout	Error! Bookmark not defined.
6.5 Evaluation & Refinement	Error! Bookmark not defined.
7. Project Requirements	10
7.1 Functional Requirements:	Error! Bookmark not defined.
7.2 Non-Functional Requirements:	Error! Bookmark not defined.
7.3 System Requirements:	Error! Bookmark not defined.
8. Timeline.....	11
9. References	12

1.Introduction

In the Gampaha District, dengue remains a critical public health challenge due to rapid urbanization and fluctuating climate conditions. While the Medical Officer of Health (MOH) collects manual data regarding mosquito density, this data is often used for historical record-keeping rather than future forecasting. Currently, there is a lack of computational models that can utilize this historical data to predict future outbreaks before they occur.

This part of the project focuses on the "intelligence" layer of the system—specifically, the development of a Machine Learning (ML) model. By acquiring and analyzing the specific mosquito density dataset from the Gampaha MOH area, we aim to identify patterns and correlations between vector density and environmental factors. Unlike the hardware phase which collects real-time data, this phase aims to process historical MOH data to train a predictive engine. This will transform raw statistical records into a forward-looking warning system, allowing health authorities to deploy resources to high-risk zones *before* an outbreak peaks.

2. Background and Literature Survey

The application of Data Science in epidemiology is becoming a standard for vector control. Globally, researchers utilize historical datasets to train algorithms that can forecast disease trends. Technologies such as Python and libraries like Scikit-Learn and TensorFlow are industry standards for building these predictive models.

However, a predictive model is only as good as its local data. While generic models exist, they often fail when applied to specific localities like Sri Lanka due to unique environmental and urban variables. Processing MOH data requires rigorous data cleaning and preprocessing techniques to handle missing values and inconsistencies. By using time-series forecasting methods (like ARIMA or LSTM neural networks), researchers can analyze temporal trends in mosquito breeding. This project leverages these established data science methodologies to analyze the specific Gampaha dataset, bridging the gap between raw government statistics and actionable computational insights.

3. Research Gap

Current research in Sri Lanka often focuses on either biological vector control (chemical fogging) or simple descriptive statistics (charts of past cases).

- **Gap 1: Lack of Localization:** Most predictive models are trained on global datasets and do not account for the specific micro-climatic conditions of the Gampaha district.
- **Gap 2: Reactive vs. Predictive:** Existing MOH reports describe *what happened* last month. There is a significant scarcity of systems that tell officials *what will happen* next month based on that data.
- **Gap 3: Data Utilization:** Valuable density data sits in manual logs or static spreadsheets without being fed into automated analysis pipelines.

This project fills these gaps by developing a localized predictive model specifically tuned to the Gampaha MOH dataset, moving beyond simple visualization to algorithmic forecasting.

4. Research Problem

The core problem is the "**Insight Lag**" in the current manual reporting system.

1. **Unused Historical Value:** The Gampaha MOH possesses rich historical data on mosquito density, but it is currently an under-utilized asset.
2. **Delayed Response:** Health officials currently rely on lagging indicators (confirmed hospital cases) to declare high-risk zones. By the time cases spike, the vector density has already been high for weeks.
3. **Complex Correlation:** It is difficult for humans to manually correlate fluctuating mosquito density with complex variables like rainfall lags and urbanization without computational aid.

We need a system that can ingest this complex dataset and output a clear, simple probability score for future outbreaks.

5. Objectives

5.1 Main Goal: To design and validate a Machine Learning model capable of analyzing historical mosquito density data from the Gampaha MOH area to predict future density trends with high accuracy.

5.2 Specific Goals:

- To acquire, clean, and preprocess the Gampaha MOH mosquito density dataset to ensure it is suitable for algorithmic training (handling missing values, normalization).
- To perform Exploratory Data Analysis (EDA) to identify key correlations between time periods and density spikes in the Gampaha region.
- To train and test multiple Machine Learning algorithms (e.g., Regression, Random Forest, or LSTM) to find the most accurate model for this specific locality.
- To generate a validation report comparing the model's predicted values against actual historical records to prove the system's reliability.

6. Methodology

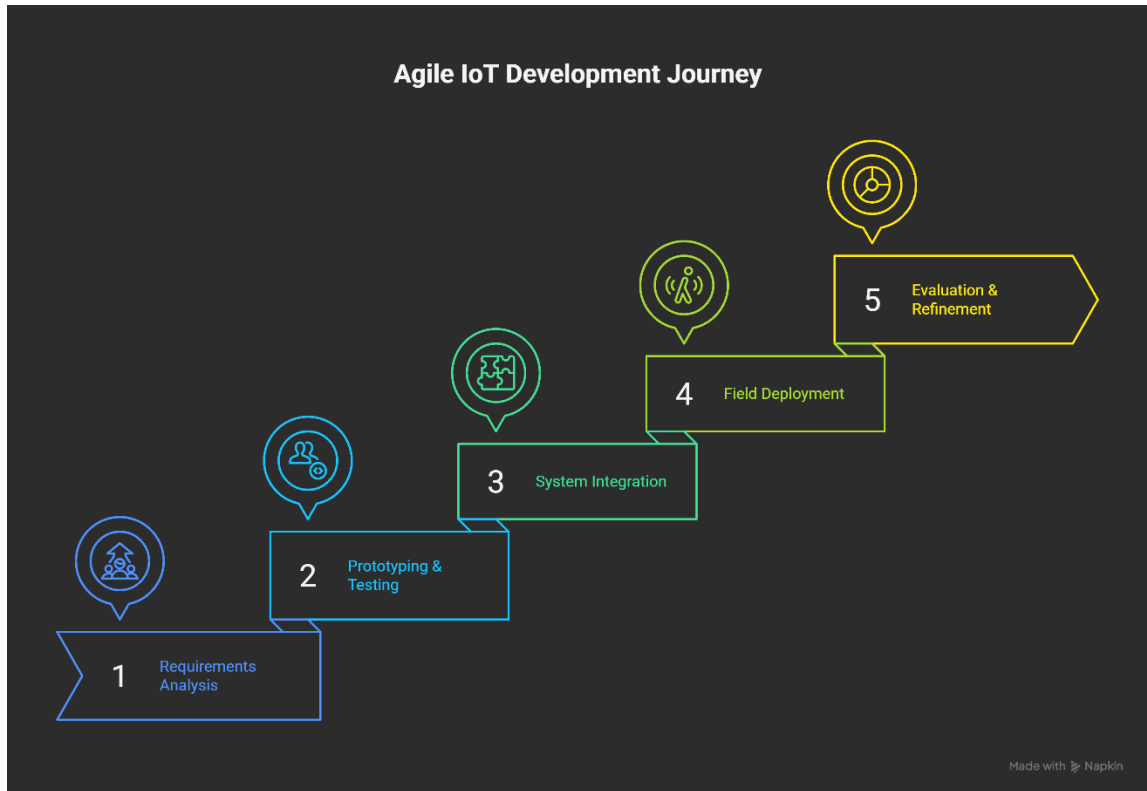
6.1 Data Acquisition & Understanding This phase involves securing the official dataset from the Gampaha MOH (via the Healthcare PD) and performing an initial audit of the data structure (columns, frequency, data types).

6.2 Data Preprocessing (Cleaning) Raw government data often contains noise. We will use Python (Pandas) to handle null values, remove duplicates, and normalize the data formats. This stage is critical to ensure the "Garbage In, Garbage Out" principle does not affect our results.

6.3 Feature Engineering & Model Selection We will identify which data points (features) have the highest impact on mosquito density. We will then select appropriate algorithms (e.g., Linear Regression for trends, LSTM for time-series) and split the data into Training and Testing sets.

6.4 Training & Evaluation The model will be trained on the training set. We will evaluate its performance using standard metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to ensure the predictions are statistically significant.

6.5 Refinement & Reporting Based on the error metrics, we will tune the model parameters (hyperparameters) to improve accuracy. The final output will be a comprehensive analysis report and a functional prediction script.



7. Project Requirements

7.1 Functional Requirements:

- The system shall ingest CSV/Excel datasets provided by the MOH.
- The system shall output a predicted density value for a future time window (e.g., next 7 days).
- The system shall visualize the correlation between predicted vs. actual values on a graph.

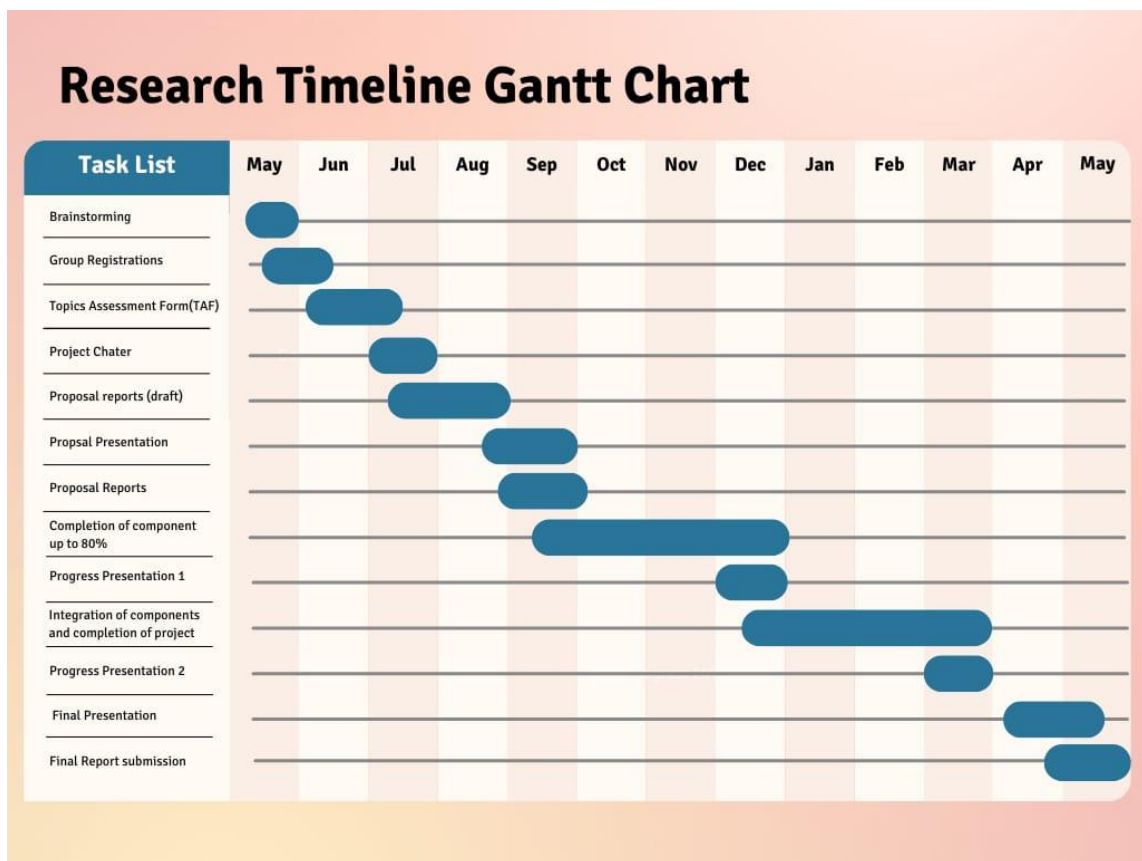
7.2 Non-Functional Requirements:

- Accuracy: The model should achieve a prediction accuracy of at least 80% on the test dataset.
- Privacy: All data processing must adhere to strict confidentiality regarding any sensitive location details in the MOH data.
- Reproducibility: The analysis script must be reproducible on standard research hardware.

7.3 System Requirements:

- Language: Python 3.9+
- Libraries: Pandas, NumPy, Scikit-Learn, Matplotlib/Seaborn
- Environment: Jupyter Notebook / Google Colab
- Data Source: Gampaha MOH Official Records (CSV/XLSX)

8. Timeline



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